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Magnetic Reversals During the Deccan Volcanism: Paleomagnetic Insights From the Pachmarhi Dykes

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Key Points:

- Fourteen dykes from the Pachmarhi dyke swarm reveal five with normal polarity, seven with reverse polarity, and two that record both polarities
- These two dykes that record both polarities provide clear evidence of a geomagnetic reversal during emplacement
- The paleopole of Pachmarhi dykes (37°N, 68.6°W) closely aligns with the Nandurbar-Dhule dykes and the Deccan Superpole

Supporting Information:

Supporting Information may be found in the online version of this article.

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Abstract The three main dyke swarms that are linked to the Deccan Continental Flood Basalts are the Nasik-Pune, Western Coastal, and Narmada-Satpura-Tapi (N-S-T) swarms. Encompassing approximately 244 mapped basaltic dykes, mainly trending E-W and positioned along an ancient tectonic zone, the Pachmarhi dyke swarm is situated in the eastern N-S-T segment. The length of these dykes varies from 140 m to 22 km, with a mean abundance between 5 and 15 km. According to petrographic and rock magnetic analyses, the main magnetic remanence carriers are high- and low-titanium magnetite particles, primarily in the pseudo-single domain state. Paleomagnetic analyses of 14 dykes show that two of them exhibit both polarities within the same dyke, seven have reverse polarity, and five have normal polarity. This implies periodic intrusion of magma during geomagnetic reversals. The mean ChRM directions for normal and reverse polarity dykes are $D_m = 329.3°$, $I_m = -40.3°$ ($k = 80.1$, $\alpha_{95} = 8.6°$, $n = 31$) and $D_m = 154.4°$, $I_m = 39.5°$ ($k = 122$, $\alpha_{95} = 5.5°$, $n = 69$), respectively. The combined mean ChRM direction ($D_m \approx 332°$, $I_m \approx -39.8°$) yields a paleopole at 37°N, 68.6°W, closely aligning with the Nandurbar-Dhule (N-D) dykes and the Deccan Superpole. The results indicate that the Pachmarhi dykes were formed during a mostly synchronous magmatic episode along the Narmada–Son Lineament. Due to the lack of direct geochronological constraints for these dykes, their precise correlation with the eruptive history of the Deccan Traps remains unresolved.

Plain Language Summary One of the largest volcanic provinces on Earth was formed by a series of massive volcanic eruptions that created the Deccan Traps ~66 million years ago. Through cracks in the earth's crust, molten magma rises upward, solidifying into dykes cutting across the lava pile. We found ~244 of these dykes in the Pachmarhi area, mostly east-west trending, in central India along the Narmada-Son lineament. These dykes once served as pathways that carried magma to the surface, helping feed the great lava outpourings that built the Deccan Traps. Similar features in the Nandurbar-Dhule regions show that these dykes were widespread, functioning as lava conduits during the Deccan volcanism. Some dykes preserve a normal magnetic field direction, while others show a reversed one, indicating that Earth's magnetic poles flipped during their formation. In this study, two of the studied dykes retained both normal and reversed magnetic signals, indicating that several magma pulses may have erupted, witnessing a geomagnetic reversal. Their estimated latitudes and recorded magnetic signals provide new evidence for the timing and dynamics of magma transport during Deccan volcanism.

1. Introduction

The Deccan Continental Flood Basalt (CFB) province, one of the largest and most well-preserved volcanic provinces on Earth, was emplaced between the Late Cretaceous and early Paleogene periods in the Indian peninsula (Baksi, 2014; Jain et al., 2020; Krishnamurthy, 2020; Mittal, Sprain, et al., 2021). Covering an area of approximately 500,000 km^2 (Watts & Cox, 1989), the Deccan Traps represent a colossal magmatic event, with a total outpouring of $\sim 1.3 \times 10^6$ km^3 of basaltic magma (Jay & Widdowson, 2008). This magmatism, associated with the Réunion hotspot, coincided with significant geodynamic and evolutionary events, such as the northward drift of the Indian Plate and major environmental and biotic changes at the K–Pg boundary (e.g., Self et al., 2006).

Dyke swarms, representing discordant mafic intrusions within the Deccan Traps, served as magma conduits and pathways for magma during volcanic events. These dykes are primarily concentrated in three major swarms (Figure 1a): the ENE-WSW trending Narmada-Satpura-Tapi (NST) dyke swarm, the randomly oriented Nasik-Pune dyke swarm, and the N-S trending Western Coastal dyke swarm (e.g., Auden, 1949; Bondre et al., 2006; Melluso et al., 1999; Vanderkluysen et al., 2011). Kale et al. (2020) further divided the NST dykes into the

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Narmada Valley dykes, which intrude the Deccan basaltic flows and Proterozoic rocks, and the Satpura dykes, which cut both Deccan basalts and Gondwana sediments.

The Pachmarhi Dyke swarm, a part of the Satpura dykes, which are studied here, marks the eastern extent of the NST dyke swarm (Figure 1a). The dykes predominantly exhibit a preferred orientation of N82°E (~E-W trending; Figure 1b; Sheth et al., 2009; Shukla et al., 2022), parallel to the Narmada-Son Lineament (NSL), and were emplaced perpendicular to the minimum horizontal stress (σ_3) direction, aligned along the ~N-S trending paleo-extension direction (e.g., Das & Mallik, 2021).

Geochemical analyses by Sheth et al. (2009) indicate that the Deccan intrusions around Pachmarhi are more evolved than many of the Deccan basalts, displaying lower magnesium numbers (Mg#) and higher TiO_2 content, likely attributed to mixing between Deccan basaltic magmas with partial melts of Precambrian amphibolites. Bouguer gravity modeling by Bhattacharji et al. (2004) identified dense mafic magma bodies at 6–8 km depths along the Narmada-Tapi intraplate rift zone, suggesting their role as primary magma reservoirs for the Deccan flood basalts. This finding was further corroborated by K. N. D. Prasad et al. (2018) through 3D density modeling, which confirmed the presence of dense magma reservoirs at depths of 4–7 km. Mittal, Richards, and Fendley (2021) proposed a multi-connected magma reservoir model, challenging the conventional single-reservoir hypothesis by suggesting interconnected trans-crustal magma chambers as sources for Deccan volcanic episodes. Aspect ratio (length/thickness) analyses of the Nandurbar-Dhule and Pachmarhi dykes constrain source depth and magmatic overpressure through fracture-mechanics relationships, supporting the existence of shallow, upper-crustal magma chambers (Ray et al., 2007; Shukla et al., 2022). Additionally, Anisotropy of Magnetic Susceptibility (AMS) studies from different Deccan dyke swarms, such as the Nandurbar-Dhule dykes (Das et al., 2021), Pachmarhi dykes (Shukla et al., 2024), and Western Coastal dykes (Lakshmi et al., 2024), showed primarily sub-vertical to inclined magma flow patterns. These studies showed that, at the regional level, magma was supplied by multiple shallow crustal chambers (also referred to as a polycentric system), while individual flow units were fed by a single magma conduit. Furthermore, paleomagnetic analysis of the Nandurbar-Dhule dykes (Das & Mallik, 2021) suggested a maximum northward drift of the Indian plate, estimated at approximately 25 ± 8 cm/year during the late phase of Deccan emplacement. These findings underscore the tectonic significance of the Deccan intrusions and provide crucial insights into the magmatic emplacement mechanisms and plate dynamics during this period.

The present study focuses on the paleomagnetic analysis of the Pachmarhi dykes to determine their Virtual Geomagnetic Pole (VGP) and paleolatitude during emplacement. Understanding the paleolatitude of the Indian Plate at the time of dyke emplacement provides essential insights into the tectonic evolution of the Deccan volcanic province and aids in reconstructing the drift history of the Indian subcontinent.

2. Geological Background and Field Studies

Pachmarhi, a town in Central India, is located within the rhomb-shaped Satpura-Gondwana pull-apart basin, which spans approximately 200 km in length and 60 km in width at an elevation of ~1,050 m above mean sea level. This basin hosts a siliciclastic stratigraphic sequence of about 5 km, ranging from the Permian to the Cretaceous period, overlying the Precambrian basement (e.g., Crookshank, 1936; Ghosh & Sarkar, 2010). The geological succession of Pachmarhi is divided into two primary groups (Figure 1c): the Lower Gondwana Group, comprising the Talchir, Barakar, Motur, and Bijori formations, and the Upper Gondwana Group, which includes the Pachmarhi, Denwa, and Bagra formations (e.g., Chakraborty & Ghosh, 2005; Ghosh & Sarkar, 2010). Post-Deccan uplift within the Satpura-Gondwana basin has elevated and warped pre-Deccan planation surfaces, forming the 160 km-long ENE-WSW trending Pachmarhi scarp at an altitude of 280 m (e.g., Sheth et al., 2009). South of Tamia, another raised scarp reveals the Gondwana succession capped by three Deccan basaltic flows, while the Chakhla-Delakhari mafic sill (PMS1; Figure 1c) intruded the succession at a lower elevation (e.g., Crookshank, 1936; Sheth et al., 2009). The Deccan dykes and sills intruding the Lower and Upper Gondwana formations near Pachmarhi predominantly exhibit basaltic compositions (e.g., Sheth et al., 2009).

The Pachmarhi dykes are sub-vertical (dip > 80°), trending primarily ENE-WSW or E-W, with a dominant orientation of N82°E (Figure 1b; Shukla et al., 2022). Statistical analysis of 244 Pachmarhi dykes by Shukla

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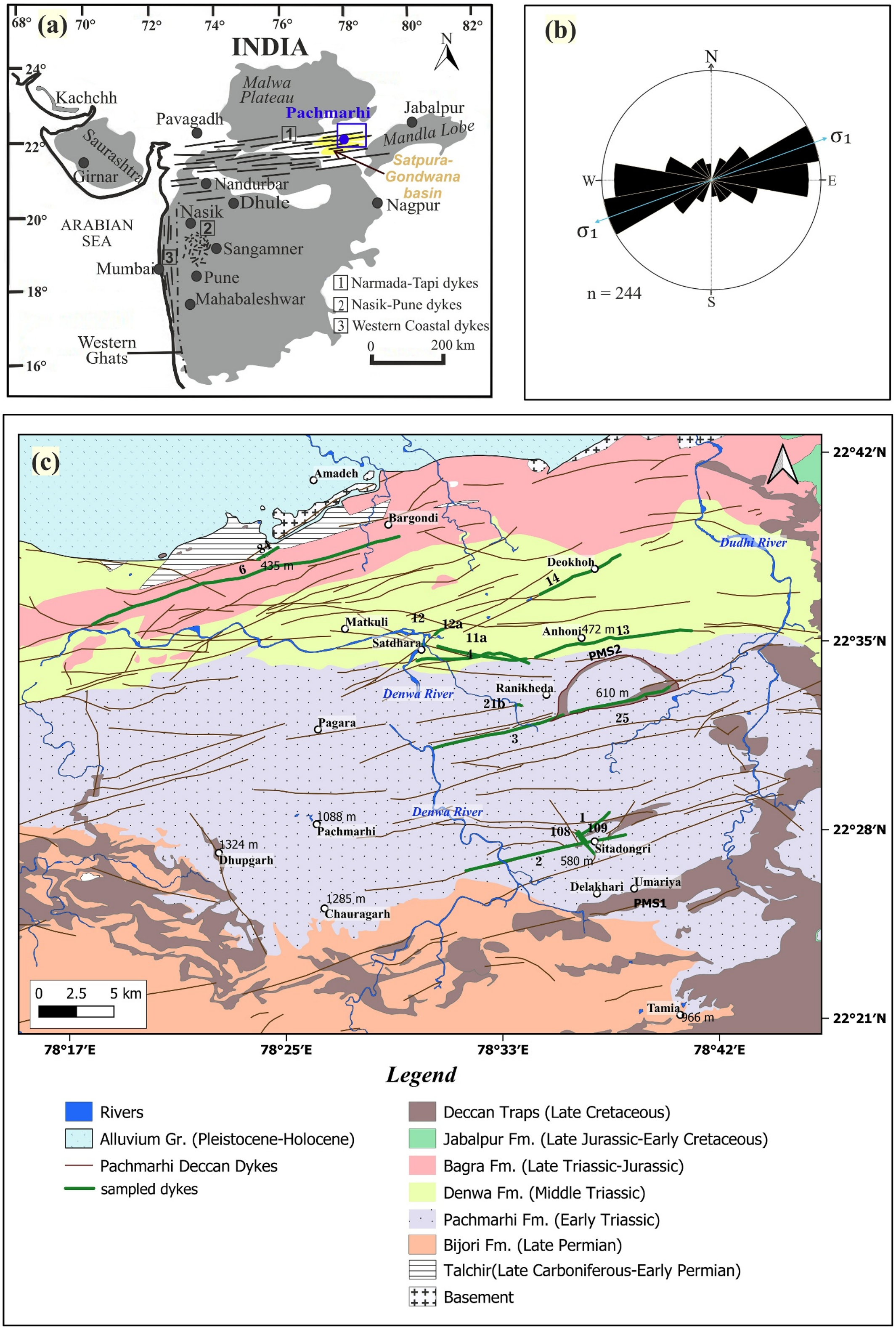

Figure 1.

et al. (2022) reported thicknesses ranging from 3.5 to 35 m (average: 13 m) and lengths from 140 m to 22 km (average: ∼5 km).

Field observations show that dykes often crop out as ridges with distinct contacts to the host rock, including sandstone and mudstone (Figures 2 and 3). From the 16 observed Pachmarhi dykes, most are weathered (e.g., PMD7a, PMD9, PMD25, and PMD84) except three dykes, for example, PMD11, PMD12, and PMD21b (Figures 2a–2c). The dykes are composed of fine- to coarse-grained basalts. Near Satdhara along the Denwa River, cross-cutting dykes suggest multiple emplacement phases, where dyke PMD12 intrudes the eastern end of PMD11 (Figures 2a and 2b). The dyke PMD6 displays multiple sub-vertical intrusions from a single dyke conduit (Figure 3a).

3. Methodology

A total of 123 oriented block samples were collected from the margins ($n = 92$) and centers ($n = 31$) of 15 dykes, covering their entire thickness. Sampling primarily followed the "H" pattern recommended by Das et al. (2021) for rock magnetic and paleomagnetic studies. Standard specimens of 2.5 cm in diameter and 2.2 cm in height were prepared by coring and cutting the oriented block samples collected from different dykes. The results of the rock magnetic and AMS analyses have been published in Shukla et al. (2024). Selected dyke sampling locations are indicated with red dots in Figures 2a, 2b, and 3a, while red flags mark locations in Figures 2c and 3b. Petrographic analysis was conducted on thin sections prepared from oriented block samples. Initial mineralogical observations were performed using transmitted light microscopy, followed by scanning electron microscopy (SEM) and energy-dispersive spectroscopy to characterize mineral phases, particularly magnetic minerals responsible for the magnetic fabric.

Magnetic property measurements were carried out using a Superconducting Quantum Interference Device Vibrating Sample Magnetometer (SQUID-VSM; MPMS-3 model, Quantum Design, USA) housed at Indian Institute of Science Education and Research (IISER) Bhopal. Temperature-dependent magnetization studies involved heating samples up to 700°C, followed by cooling to room temperature, with the Curie temperature (Tc) determined from the peak of the second derivative curve, smoothed using the cubic-spline method (Leonhardt, 2006). Hysteresis loops were obtained by applying a maximum field of 7,000 mT, and key magnetic parameters, including saturation remanent magnetization (M_{rs}), saturation magnetization (M_s), coercivity remanence (H_{cr}), and coercive force (H_c), were computed using RockMagAnalyzer 1.0 software.

In the case of dykes, the characteristic remanent magnetization (ChRM) is of thermal origin that has been acquired during cooling. However, various post-emplacement components of magnetization comprise the resultant natural remanent magnetization (NRM). The primary objective of paleomagnetic laboratory work is to isolate the ChRM. To achieve this, progressive and stepwise demagnetization techniques were conducted using both thermal (TH) and alternating field (AF) demagnetization techniques to isolate the primary ChRM directions and remove potential secondary overprints. For paleomagnetic analysis, the NRM of the specimens was initially measured using a Portable Spinning Magnetometer (PSM, MAG Instruments), which has a sensitivity of $<1 \times 10^{-9}$ Am^2 (1×10^{-4} A/m).

Stepwise AF demagnetization was performed at increasing field strengths of 2.5, 5, 7.5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 60, 80, and 100 mT using a Molspin AF demagnetizer (ASC Scientific D-2000 AF Demagnetizer). Thermal demagnetization was carried out in temperature increments of 100, 150, 200, 250, 300, 350, 400, 450, 500, 550, and 580°C using an MMTD-80 thermal demagnetizer (Magnetic Measurements, UK).

The Zijderveld diagram (Zijderveld, 1967) was used to plot the demagnetization data using the Principal Component Analysis (Kirschvink, 1980). The site mean directions were calculated using Fisher's statistics

Figure 1. (a) A map of Central-Western India illustrating the locations of the three major dyke swarms along with the present-day extent of the Deccan flood basalts. The dyke trends and locations of the major dyke swarms are also indicated. (b) A rose diagram representing the orientation of the maximum principal stress (σ_1) for 244 dykes, adapted from Shukla et al. (2022). (c) A detailed map of the Pachmarhi dyke swarm, showing lithologies, geographic features, localities, and dyke distributions (modified from Sheth et al., 2009; Shukla et al., 2022). Dyke numbers correspond to sampled locations and are prefixed as PMD, following Sheth et al. (2009), though the prefix is not displayed on the map. The sampled dykes are marked with green solid lines, whereas the rest of the dykes are marked with solid brown lines. Elevation is provided in meters. Additionally, PMS1 represents the Chakhla-Delakhari mafic sill, while PMS2 corresponds to a saucer-shaped sill (Sheth et al., 2009).

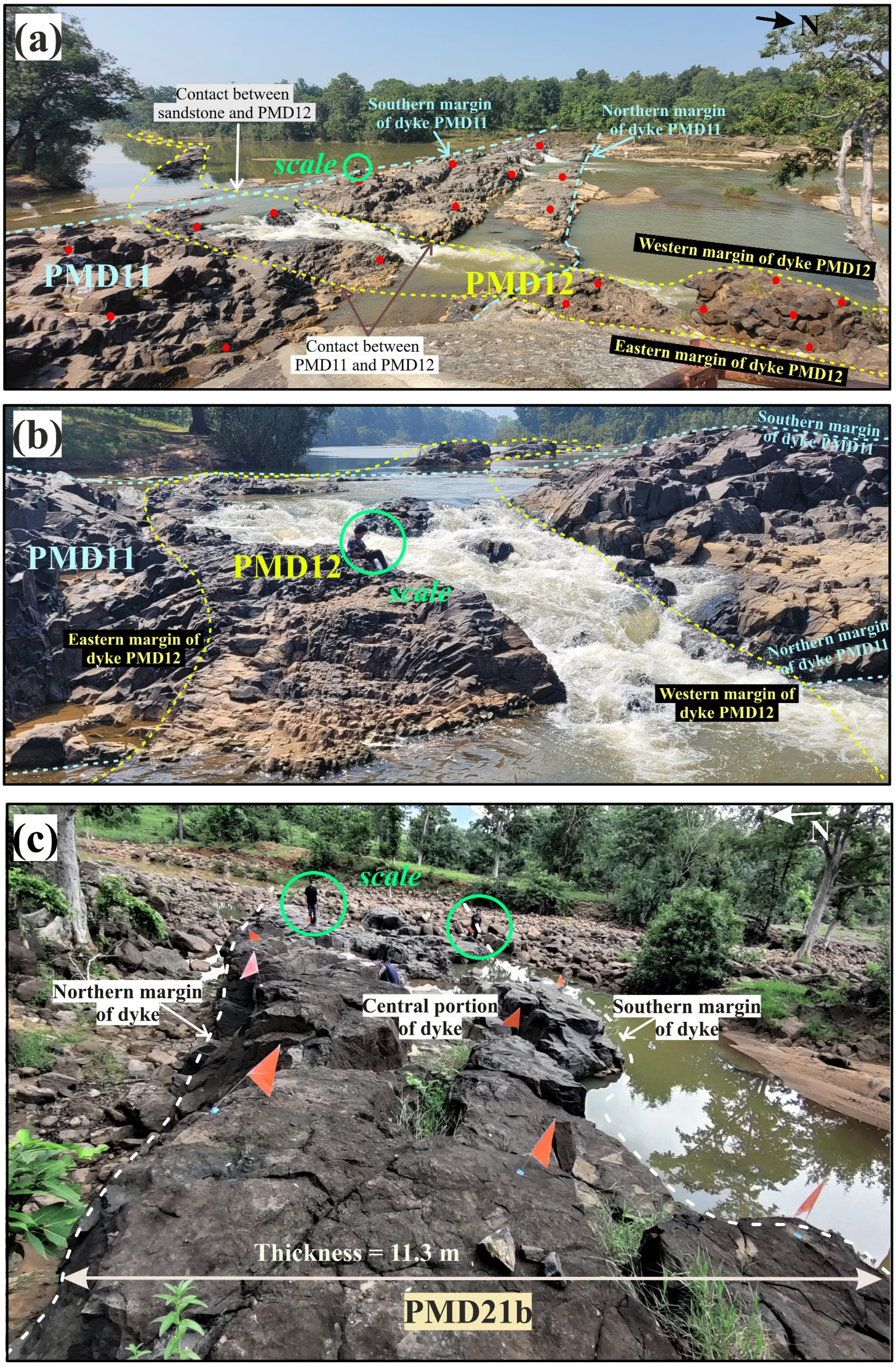

Figure 2. Field photographs: (a) Dyke PMD12 (average thickness: 15 m) cross-cutting dyke PMD11 (average thickness: 27.4 m) near the Satdhara viewpoint on the Denwa River bed. The contact between PMD11 and the host rock (Upper Gondwana sandstone), as well as the contact between PMD11 and PMD12, is marked with distinct solid-dashed lines. Red dots indicate the sampling locations for both dykes. (b) A closer view of the contact between dykes PMD11 and PMD12. (c) Dyke PMD21b (average thickness: 11.3 m), with sampling locations marked by red flags.

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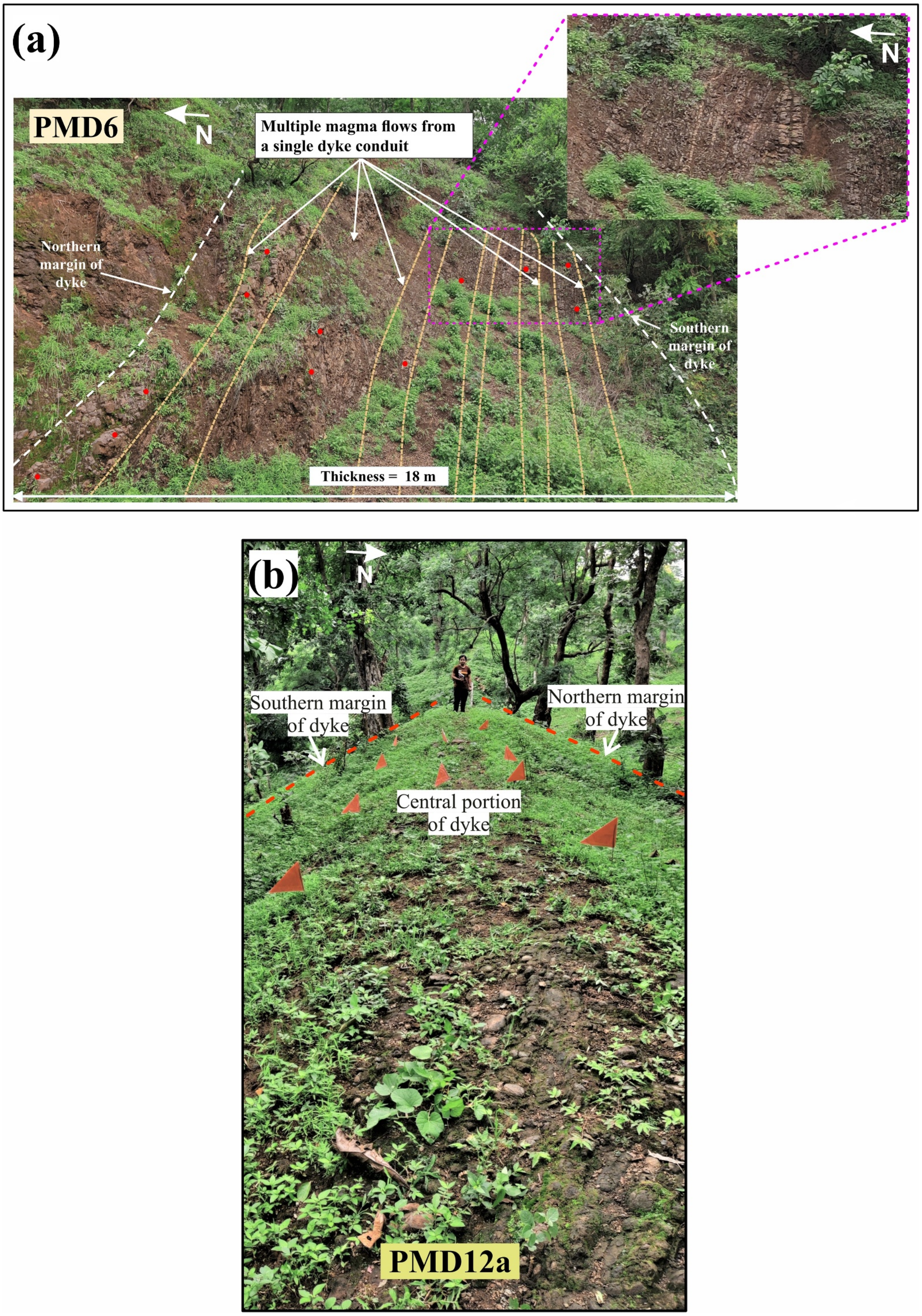

Figure 3. (a) PMD6 (thickness: 18 m) exhibits multiple magma flow units emerging from a single dyke conduit. The dyke is nearly vertical in the field, though the photograph is taken from an angled perspective. Red dots indicate sampling locations, while the zoomed-in section highlights multiple flow units. (b) PMD12a (thickness: 9 m), with sampling locations marked by red flags.

(Fisher, 1953) and the PMTools application (Efremov & Veselovskiy, 2023). All the paleomagnetic experiments were conducted at the environmental magnetism laboratory of the Indian Institute of Geomagnetism (IIG), Navi Mumbai, India.

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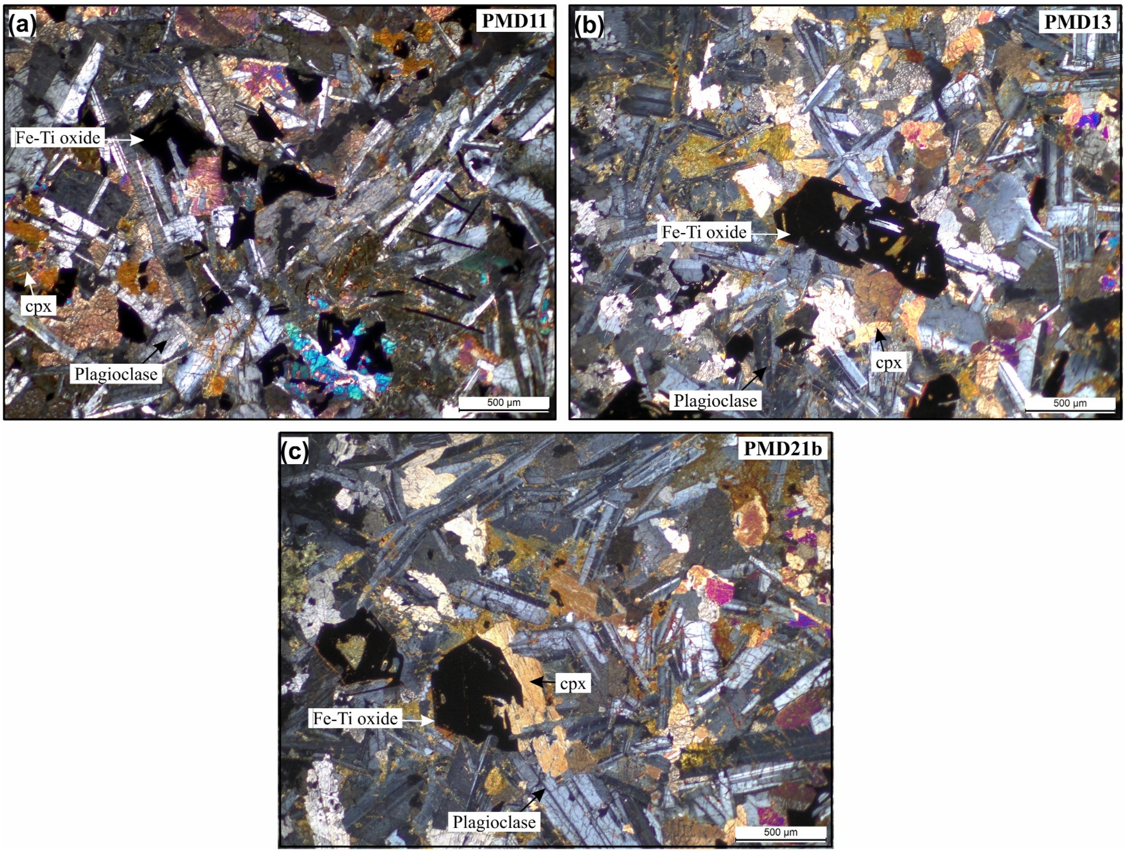

Figure 4. Photomicrographs of dyke samples PMD11, PMD13, and PMD21b under cross-polarized light, highlighting different mineralogical textures. (a) PMD11 displays an intergranular texture with Fe–Ti oxides, clinopyroxene (cpx), and plagioclase laths forming a well-developed framework. The sub-ophitic texture is also observed. (b) PMD13 exhibits a subophitic texture, where clinopyroxene partially encloses plagioclase. (c) PMD21b shows an intergranular to porphyritic texture, characterized by larger hexagonal Fe–Ti oxide grains, large plagioclase laths, and clinopyroxene forming a crystalline network.

4. Results

4.1. Petrographic Analysis

The dyke samples vary from fine to coarse-grained basalt. These rocks are mostly crystalline with a negligible amount of glass. Petrographic examination reveals that the dykes primarily consist of plagioclase laths, clinopyroxene (cpx), and Fe–Ti oxides (Figure 4). Plagioclase exhibits Carlsbad and albite twinning. In contrast, clinopyroxene appears as interstitial grains between plagioclase laths. The texture varies among samples, displaying porphyritic, subophitic to intergranular texture, where pyroxene partially encloses plagioclase.

The thin-section images of basalt samples PMD11, PMD13, and PMD21b contain variable morphologies of the Fe–Ti oxides, which are indicative of different cooling histories and crystallization environments. Magnetite and hematite are some of the most important oxides in the paleomagnetic record, retaining the Earth's magnetism during the time of their formation. Differentiation in mineralogical and textural characteristics among PMD11, PMD13, and PMD21b reflects variation in cooling rates and crystallization histories within the dykes. In PMD11 (Figure 4a), Fe–Ti oxides occur as irregular to subhedral grains, distributed along the grain boundaries of plagioclase and clinopyroxene, suggesting an early stage crystallization from the melt. Some oxide grains show elongated or needle-like forms, possibly indicating rapid cooling and oxidation. In PMD13 (Figure 4b), Fe–Ti oxides appear as larger, well-developed subhedral to hexagonal grains, implying a slower cooling rate that allowed for better crystal growth. These oxides are surrounded by a fine-grained groundmass of plagioclase and clinopyroxene, with some displaying high relief and opaque textures, characteristic of titanomagnetite and ilmenite, which are the primary carriers of remnant magnetization. In PMD21b (Figure 4c), Fe–Ti oxides occur as large hexagonal grains, embedded within a crystalline matrix of plagioclase and clinopyroxene. The presence of

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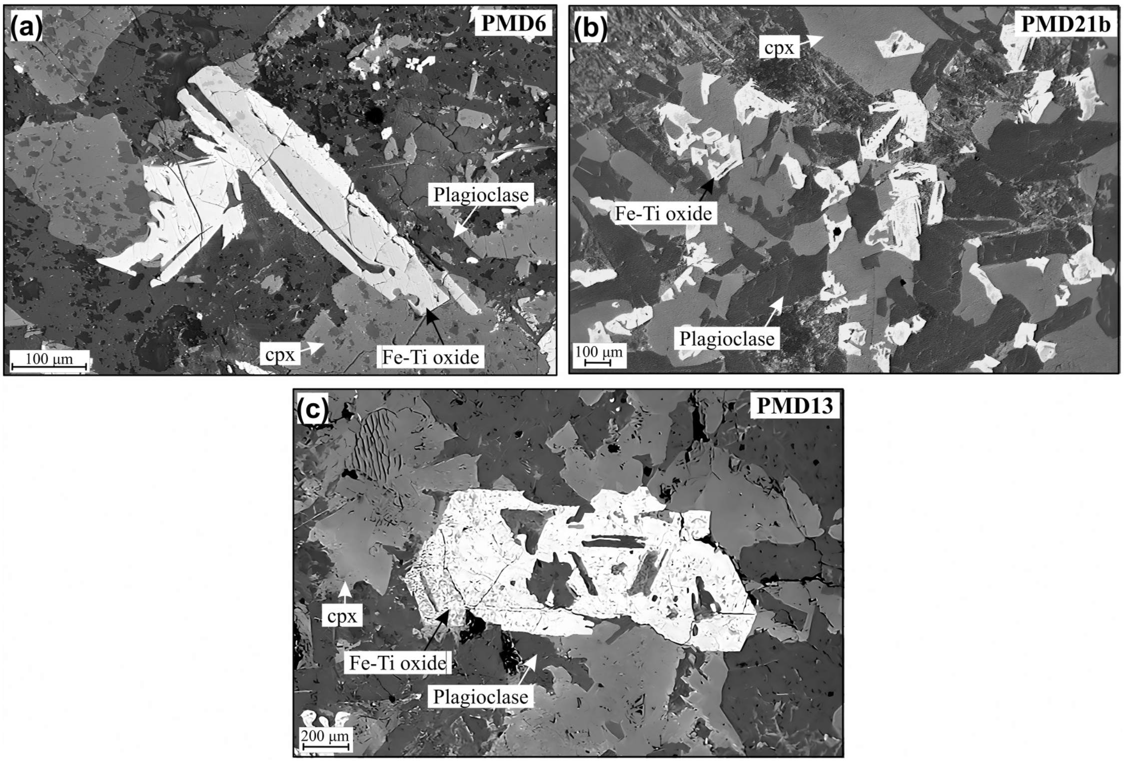

Figure 5. (a–c) Backscattered electron (BSE) images of Pachmarhi dyke samples (PMD6, PMD21b, and PMD13) obtained using SEM-EDS analysis.

hexagonal Fe–Ti oxides suggests a more prolonged cooling history, whereas the needle-shaped grains indicate faster crystallization in some portions of the rock. These oxides serve as the primary carriers of remanent magnetization in the dykes, preserving magnetic signatures that provide critical insights into the magmatic and thermal evolution of the region.

4.2. SEM-EDS Analysis

Fe–Ti oxide composition, distribution, and morphology in the studied basaltic dykes were examined using SEM-EDS analysis. Figure 5 shows the backscattered electron (BSE) images. The findings indicate that Fe–Ti oxides are found as euhedral to subhedral grains, mainly occupying the interstitial spaces between clinopyroxene and plagioclase. They display a wide range of grain sizes (~50–500 μm) and morphologies, reflecting variable crystallization conditions and cooling rates. In PMD6 (Figure 5a), a large plagioclase lath is parallel to the surrounding grains, which indicates a crystalline framework that formed during slow cooling. The Fe–Ti oxide grain is partially intergrown with pyroxene, indicating either simultaneous crystallization or late-stage melt interaction (Haggerty, 1991). In PMD21b (Figure 5b), Fe–Ti oxides form a skeletal texture that fills the interstices between plagioclase and clinopyroxene. This texture indicates rapid cooling, which limits crystal growth (Frost & Lindsley, 1991). The skeletal texture suggests that Fe–Ti oxides crystallized from the remaining iron-rich melt (Haggerty, 1991). In PMD13 (Figure 5c), Fe–Ti oxides are euhedral grains that enclose plagioclase laths, which indicates that Fe–Ti oxides continued to crystallize after plagioclase formation, suggesting a prolonged cooling history. The enclosure of plagioclase within Fe–Ti oxides implies that plagioclase crystallized in an early phase, while Fe–Ti oxides grew later. This suggests late-stage Fe–Ti oxide crystallization, likely driven by melt differentiation, where Fe and Ti became concentrated in the residual melt, eventually forming oxides around pre-existing minerals. These observations highlight the thermal and magmatic evolution of the dykes, with Fe–Ti oxides serving as key indicators of their cooling histories. No clear evidence of deuteric oxidation, such as exsolution lamellae or trellis textures, was observed in the BSE images. However, the grain sizes imaged here are generally too coarse to represent the actual carriers of stable remanent magnetization, which are likely much finer titanomagnetite grains not resolvable by SEM. These textural observations, together with rock magnetic data (M–T curves), support titanomagnetite in varying oxidation states as the dominant magnetic carrier.

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Furthermore, the morphology and composition of Fe–Ti oxides affect their capacity to acquire and retain remanent magnetization, which has a direct effect on the accuracy of paleomagnetic data, making the SEM-EDS analysis essential for paleomagnetic work.

4.3. Rock Magnetic Analysis

The rock magnetic analysis of the Pachmarhi dykes reveals variations in the magnetic mineralogy and domain states of Fe–Ti oxides. The temperature-dependent magnetization (M-T) analysis shows that the Curie temperature (T_c) ranges from 440°C (PMD11) to 563°C (PMD9), indicating the presence of titanomagnetite with varying titanium content. For PMD21b, the T_c is ∼530°C (Figure 6a), confirming titanomagnetite as the primary magnetic carrier. In the sample (e.g., PMD21b, Figure 6a), an additional inflection near 300°C is observed, which is attributed to titanomaghemite, formed by partial oxidation of titanomagnetite. The irreversible nature of these curves (more M-T curves are shown in Shukla et al., 2024) supports this interpretation. By contrast, the higher-temperature drops between 530 and 555°C correspond to low-Ti titanomagnetite or moderately oxidized titanomagnetite. Overall, titanomagnetite is identified as the dominant magnetic phase in the Pachmarhi dykes, consistent with earlier rock-magnetic observations from the dyke swarm (Shukla et al., 2024).

The hysteresis measurements (M–H) on the dyke samples were used to calculate the ratios of M_{rs}/M_s (saturation remanent magnetization to saturation magnetization) versus H_{cr}/H_c (remanent coercivity to coercivity), which were then plotted on the Day–Dunlop plot (after Day et al., 1977; Dunlop, 2002; and is given in Fig. 7d in Shukla et al., 2024). A representative sample from the PMD21b dyke (Figure 6b) shows a thin hysteresis loop, consistent with pseudo-single-domain (PSD)-like behavior. The coercivity (H_c) is 27.02 mT, and the remanent coercivity (H_{cr}) is 98.18 mT. Across all dykes, coercivity remanence ranges from 3.4 to 100.52 mT. Based on the Day–Dunlop plot (Shukla et al., 2024), 72.22% of the samples display PSD-like behavior, while 11.11% exhibit multi-domain (MD)-like behavior. These observations indicate that titanomagnetite is the principal remanence carrier in the Pachmarhi dykes, with mineralogical variations reflecting differences in cooling rates and oxidation histories. The presence of both PSD and MD grains suggests a potential for stable remanent magnetization, making these dykes suitable for paleomagnetic studies.

4.4. Paleomagnetic Analysis

Of the 136 specimens selected for paleomagnetic analysis, 20 core specimens were discarded due to ambiguous scattering because of unstable ChRM. The AF and TH demagnetization results were analyzed using Zijderveld diagrams (Zijderveld, 1967), equal-area projections, and intensity decay curves to track the progressive removal of secondary magnetization components (Figures 7 and 8).

Figure 7 illustrates the AF demagnetization results for representative dyke samples PMD11, PMD12, and PMD14. The Zijderveld diagrams illustrate a systematic decay of NRM with increasing AF steps. The demagnetization trajectories show a stable linear decay toward the origin, indicating a well-defined ChRM. The equal-area stereo plots further confirm that the remanence vectors cluster tightly as the higher coercivity components are progressively removed. The normalized NRM decay curves or intensity decay curves for AF reveal a rapid initial decline at lower coercivity steps, followed by a gradual decrease, indicating the presence of a dominant low-to-intermediate coercivity component. Additionally, the intensity decay curves for some specimens show a smooth decrease in intensity, with a reduction of approximately 50% after the 50 mT step.

Figure 8 illustrates the TH demagnetization results for representative dyke samples PMD11, PMD12, and PMD14. The specimens analyzed for TH were taken from the same samples as those subjected to alternating field (AF) demagnetization to assess the consistency in vector behavior during the demagnetization process. Stepwise heating was conducted up to 580°C, which indicates the thermal unblocking temperatures of magnetic minerals, primarily magnetite. The Zijderveld diagrams show a systematic decay of NRM with increasing temperature, demonstrating the stepwise removal of secondary overprints. Equal-area stereonets reveal the directional changes in remanence, with data points converging toward a stable ChRM direction as higher-temperature components are isolated. The intensity decay curves (Figure 8) show that during heating, most samples exhibit a significant decline in remanence around ∼450°C, followed by a smooth decline to saturation below 580°C. ChRM directions were obtained, considering the steps from 350 to 580°C for thermal and 20–100 mT for AF. Results from AF and TH demagnetization show that specimens from dykes PMD2, PMD11, PMD12, PMD12a, PMD21b, PMD25, and

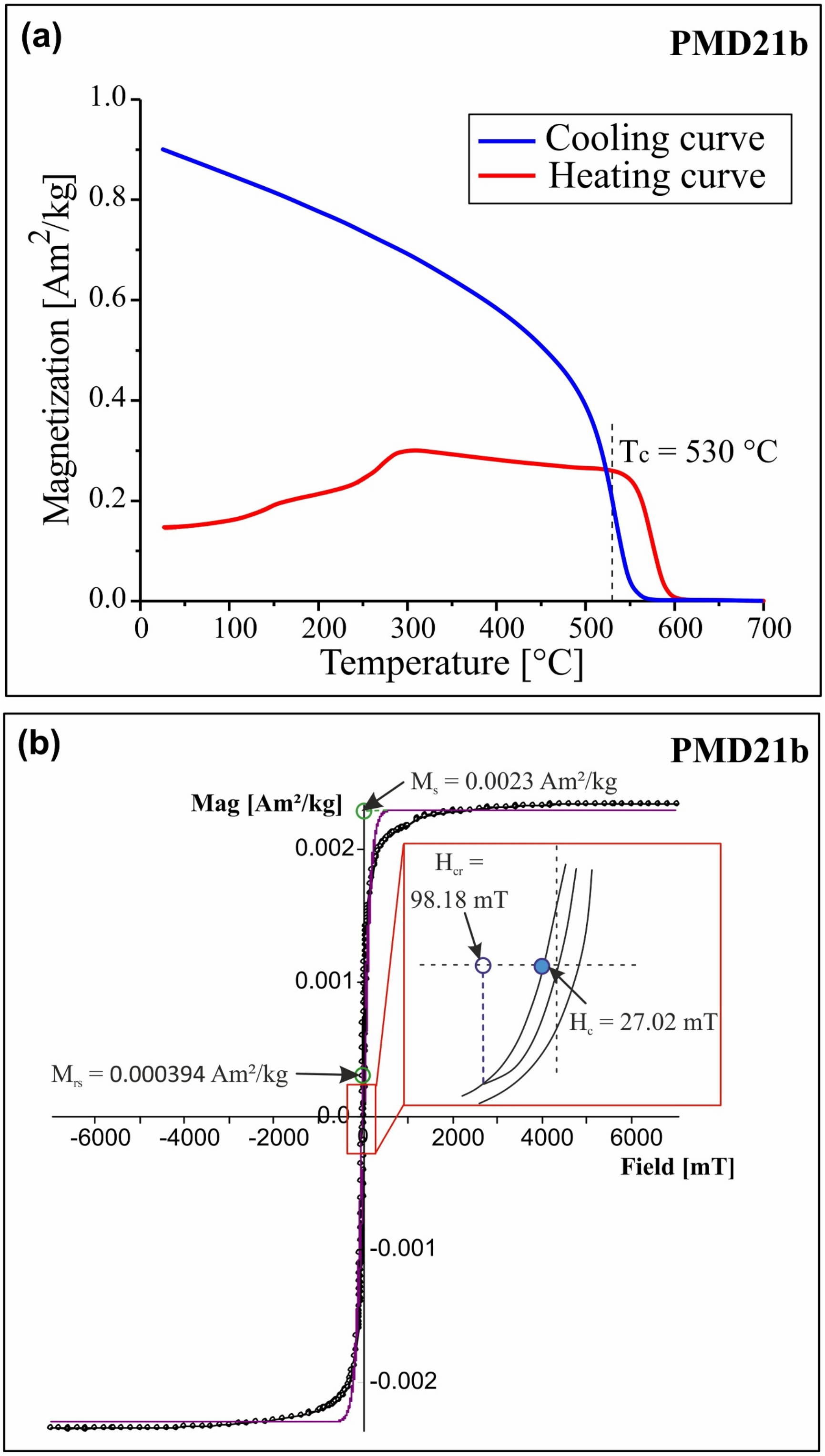

Figure 6. Rock magnetic properties of PMD21b. (a) Thermomagnetic curves indicating a Curie temperature (~530°C) and characteristics of Fe–Ti oxides (modified after Shukla et al., 2024). (b) Hysteresis loop obtained after removing paramagnetic and diamagnetic contributions (after Leonhardt, 2006). The hysteresis parameters, saturation remanence magnetization (M_{rs}), saturation magnetization (M_s), coercivity (H_c), and coercivity of remanence (H_{cr}) indicate pseudo-single domain behavior (see Figure 7d in Shukla et al., 2024).

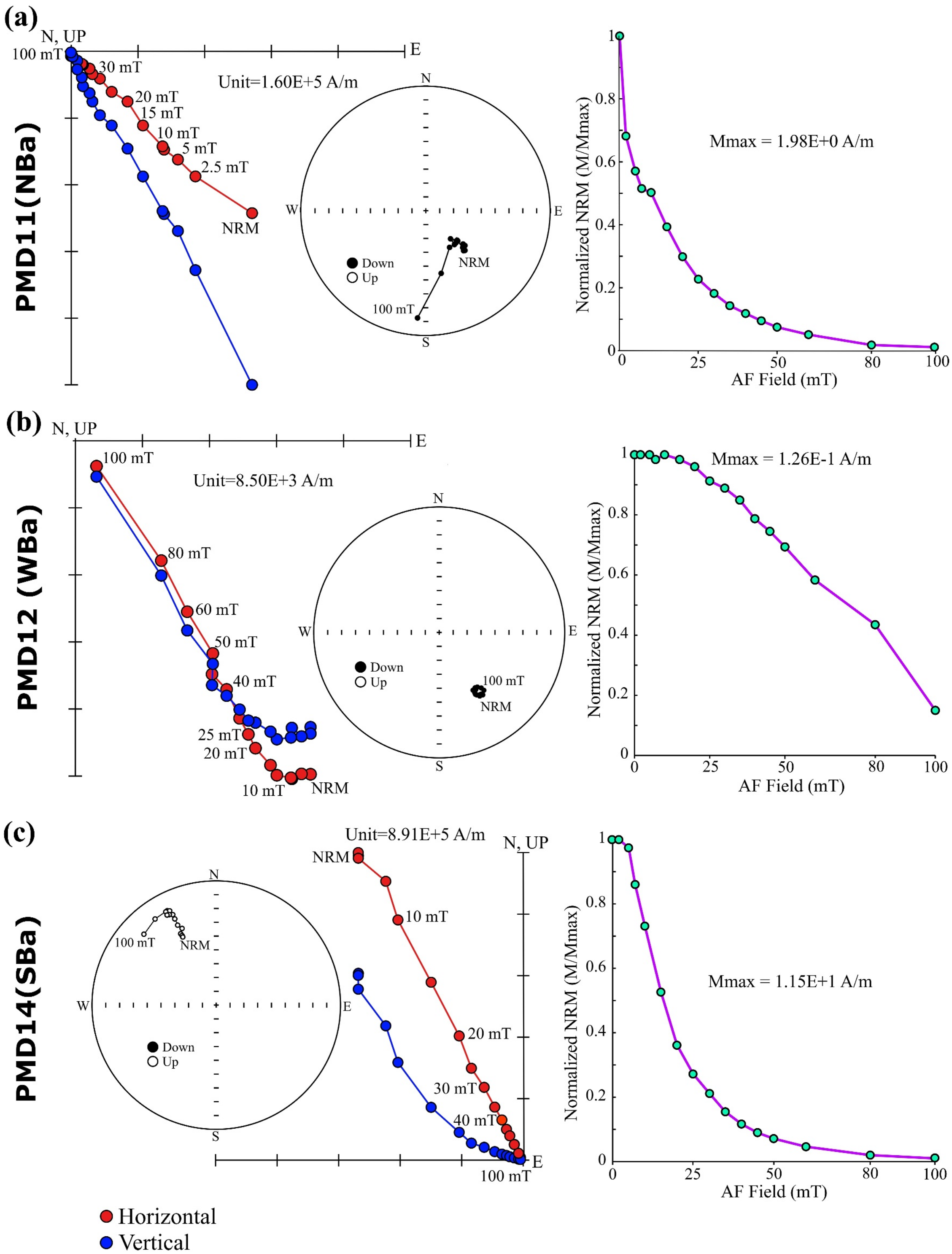

Figure 7. (a–c) Representative alternate field demagnetization results of Pachmarhi dykes specimens PMD11 (a), PMD12 (b), and PMD14 (c); other examples are shown in the Supporting Information S1 (Figure S1). Orthogonal (Zijderveld) vector plots of representative demagnetization behavior of specimens from the Pachmarhi dykes (no tilt corrections need to be applied on the samples from essentially vertical dykes). Equal-area stereonets show solid (open) symbols projected on the lower (upper) hemisphere at successive demagnetization steps. Remanence intensity decay curves are provided for each alternating field step. In the orthogonal plots, red, and blue symbols represent projections onto the horizontal and vertical planes, respectively. NRM refers to Natural remanent magnetization.

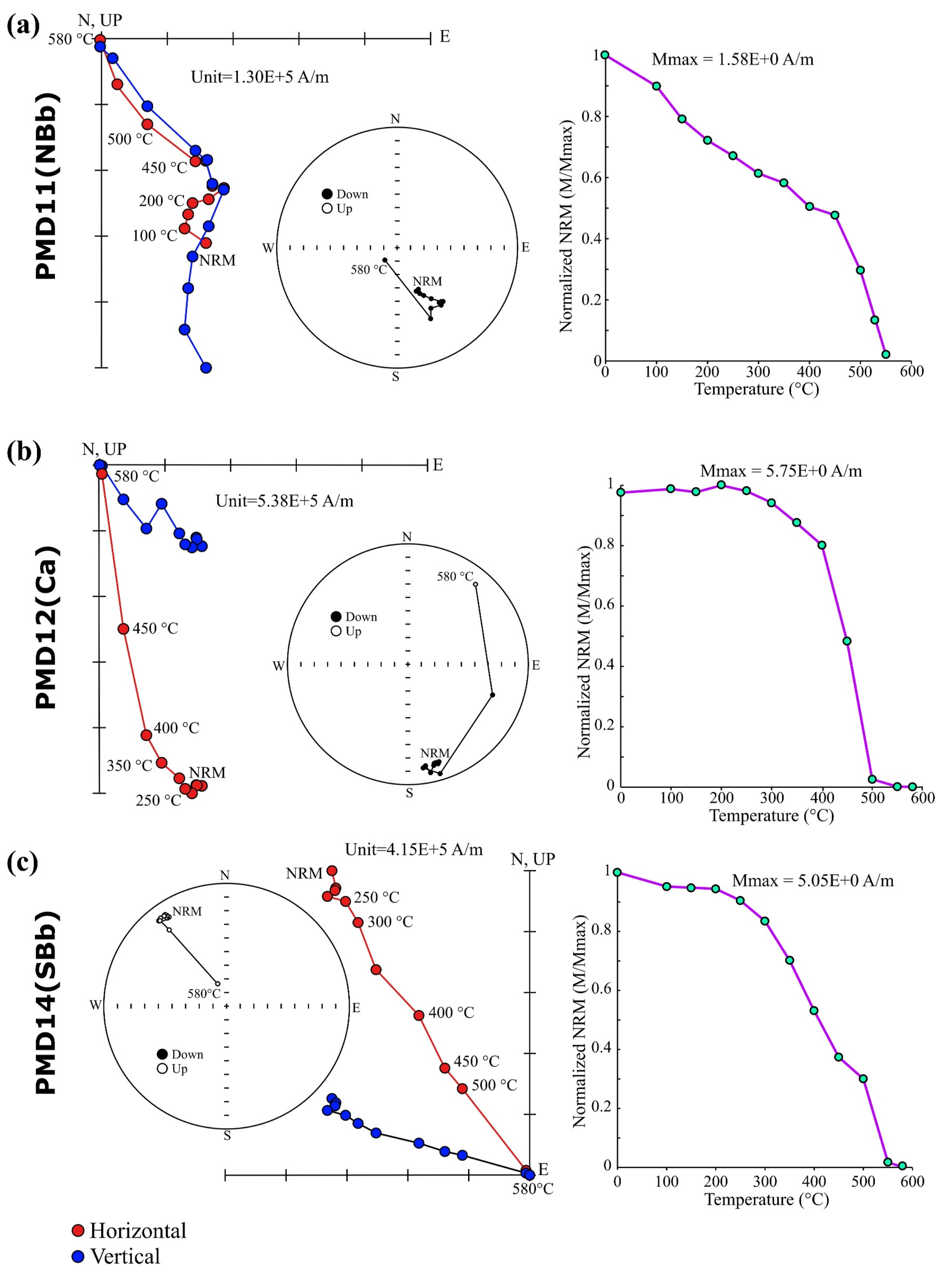

Figure 8. (a–c) Representative thermal demagnetization results of Pachmarhi dykes specimens PMD11 (a), PMD12 (b), and PMD14 (c); other examples are shown in the Supporting Information S1 (Figure S2). Orthogonal (Zijderveld) vector plots of representative demagnetization behavior of specimens from the Pachmarhi dykes (no tilt corrections need to be applied on the samples from essentially vertical dykes). Equal-area stereonets show solid (open) symbols projected on the lower (upper) hemisphere at successive demagnetization steps. Remanence intensity decay curves are provided for each TH step. In the orthogonal plots, red and blue symbols represent projections onto the horizontal and vertical planes, respectively. NRM refers to Natural remanent magnetization.

Table 1
Paleomagnetism Results From the Pachmarhi Dykes

Dyke (prefix PMD)	ChRM (AF + thermal) D_m (°)	I_m (°)	N	n	k	α_{95} (°)	Polarity	VGP (raw) Lat	Long (°E)	VGP (corrected) Lat (°N)	Long (°E)	λ_p (°S)
2	160	29.9	11	11	106.7	4.4	*R*	−46.81	107.32	46.81	287.32	16
11	150	39.9	13	13	51	5.9	*R*	−36.13	113.33	36.13	293.33	22.7
12	156.6	32.8	13	11	74.1	5.3	*R*	−43.54	109.93	43.54	289.93	17.9
12a	151.8	42.6	8	7	31.5	10.9	*R*	−35.37	110.28	35.37	290.28	24.7
21b	149.9	50.1	9	8	59.7	7.2	*R*	−29.28	108.13	29.28	288.13	30.9
25	153.7	42.5	10	10	39.8	7.7	*R*	−36.38	108.63	36.38	288.63	24.6
109	156.8	38	11	9	146.9	4.3	*R*	−40.71	107.56	40.71	287.56	21.3
Mean values for Reverse polarity dykes (=7)	154.4	39.5	75	69	122	5.5	*R*	−38.33	109.35	38.33	289.35 A_{95} = 4.52	
3	330.3	−39.5	8	8	83.9	6.1	*N*	36.57	293.34	36.57	293.34	22.4
4	337.5	−42.6	7	6	375.4	3.5	*N*	37.93	284.68	37.93	284.68	24.7
13	316.4	−48.5	8	6	200.1	4.7	*N*	23.15	299.37	23.15	299.37	29.5
14	323.1	−37.8	10	5	77	8.8	*N*	35.65	301.15	35.65	301.15	21.2
84	335.9	−31.6	8	6	171	5.1	*N*	43.8	291.13	43.8	291.13	17.1
Mean values for Normal polarity dykes (=5)	329.3	−40.3	41	31	80.1	8.6	*N*	35.10	294.13 A_{95} = 8.81	35.10	294.13 A_{95} = 8.81	
Mean of Normal and Reverse polarity dykes (=12)	332.3 (N = 12 dykes)	−39.8	116	100	103	4.3				37	291.4 A_{95} = 4.07	22.75

Reversal test (McFadden & McElhinny, 1990):

Polarity	D_m	I_m	n	k	α_{95} (°)	Angular difference (°)	Reversal test results
Normal (*N*)	329.3	−40.3	31	80.1	8.6	~4°–5°	Positive (antipodal within scatter, passes at ~95% confidence level)
Reverse (*R*)	154.4	39.5	69	122	5.5		
Flipped Reverse	334.4	−39.5	–	–	–		

Note. N = total number of specimens measured (excluding those with ambiguous scattering); n = number of specimens used for site-mean ChRM calculation; D_m = mean declination; I_m = mean inclination; α_{95} = 95% confidence circle around the site-mean direction; k = precision parameter; λ_p = paleolatitude calculated from the site-mean inclination; Lat = Latitude of the VGP, Long = longitude of the VGP (°E); A_{95} = 95% confidence circle of the mean VGP.

PMD109 exhibit southeast (SE) declinations with positive inclinations, while specimens from PMD3, PMD4, PMD13, PMD14, and PMD84 display northwest (NW) declinations with negative inclinations (Figures 7 and 8).

The thermal demagnetization data align with the AF demagnetization data, indicating normal polarity in 5 dykes and reverse polarity in 7 dykes. AF demagnetization demonstrates a notable improvement in the clustering of sample and site directions compared to thermal demagnetization. The results from both thermal and AF demagnetization are integrated to derive an estimated mean for the characteristic remanent magnetization (ChRM) directions, ensuring statistical reliability for calculating the virtual geomagnetic pole (VGP) for individual dykes. Fisher statistics (Fisher, 1953) were used to calculate the site and group mean directions. The site mean directions were accepted for further analysis if their 95% confidence circle (α_{95}) was less than 15.

Table 1 summarizes the paleomagnetic results for the Pachmarhi dykes within the study area. The estimated mean ChRM directions for the analyzed dykes are illustrated in Figure 9. Dykes exhibiting reverse polarity have a mean ChRM direction, where mean declination (D_m) = 154.4° and mean inclination (I_m) = 39.5° (k = 122, α_{95} = 5.5°, n = 69), while those with normal polarity show a mean ChRM direction of D_m = 329.3° and I_m = −40.3° (k = 80.1, α_{95} = 8.6°, n = 31). The combined estimated $D_m \approx 332°$ and $I_m \approx -39.8°$ (k = 103, α_{95} = 4.3°, n = 100). As all of the dykes have vertical margins, no tectonic tilt correction was applied. To further assess the primary nature of the remanence, we applied the reversal test of McFadden and McElhinny (1990) using the mean

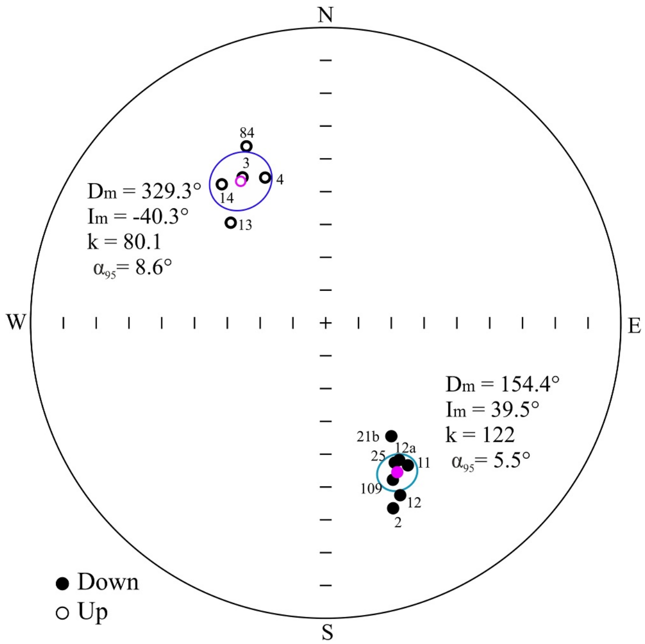

Figure 9. Equal area projections of site-mean characteristic remanent magnetization (ChRM) directions of the Pachmarhi dyke (dyke names are mentioned without the prefix PMD). Mean directions (D_m, I_m) for normal and reversed dykes are calculated separately. The other paleomagnetic parameters are given in Table 1.

normal and reversed site directions. The normal mean direction ($D_m = 329.3°$, $I_m = -40.3°$) and the antipodally corrected reversed mean direction ($D_m = 334.4°$, $I_m = -39.5°$) are extremely close, with an angular separation of ~4° (≈5° in rounded terms). This separation is significantly smaller than the critical angle at the 95% confidence level based on n, k, and α_{95}, indicating that the two populations are statistically antipodal. Thus, the reversal test is positive, supporting that the ChRMs reflect a primary magnetization acquired in a geocentric axial dipole field during the emplacement of the Pachmarhi dykes (Table 1).

5. Discussion

The paleomagnetic findings from the Pachmarhi dykes offer significant insights into their emplacement history and the wider regional tectonics. The obtained VGPs align with the anticipated position of the Indian Plate during Deccan volcanism, supporting northward movement during that period. The calculated paleopole (Table 1) for the Pachmarhi dykes (37°N, 68.6°W) closely aligns with the paleopole of the Nandurbar-Dhule (N-D) dykes (38.3°N, 79.3°W), resulting in an average position of 37.8°N, 73.9°W. This paleopole corresponds with the DSP (36.96°N, 78.7°W; Vandamme et al., 1991), highlighting the synchronous emplacement of these dyke swarms and their connection to Deccan volcanism.

Out of the 14 studied dykes, five exhibit normal magnetic polarity, while seven display reverse polarity. The remaining two dykes, PMD6 and PMD108, show magnetic reversals within the same dyke. The ChRM directions for PMD6 are illustrated in Figure 10a, with five specimens exhibiting reverse polarity ($D_m = 152.2°$, $I_m = 44.6°$, $k = 48.7$, $\alpha_{95} = 11.1°$) and four specimens showing normal polarity ($D_m = 307.1°$, $I_m = -41.2°$, $k = 57.4$, $\alpha_{95} = 12.2°$). Similarly, PMD108 also displays magnetic reversal (Figure 10b), with four reverse-polarity specimens ($D_m = 142.3°$, $I_m = 38.5°$, $k = 55.5$, and $\alpha_{95} = 12.4°$) and four normal-polarity specimens ($D_m = 310.7°$, $I_m = -28.9°$, $k = 175.5$, and $\alpha_{95} = 7°$). Given the presence of mixed polarity in PMD6 and PMD108, a paleopole was not determined for these two dykes, while the paleopoles for the remaining 12 dykes are presented in Table 1.

The calculated paleopole for the Pachmarhi dyke swarm, located at 37°N, 68.6°W (291.4°E), was compared with that of the N-D dyke swarm, which, together with the Pachmarhi dykes, forms part of the Narmada-Satpura-Tapi (N-S-T) dyke system. The paleopole of the N-D dykes is reported at 38.3°N, 79.3°W (280.7°E) (J. N. Prasad et al., 1996; Sethna et al., 1999; Das et al., 2021). This comparison confirms their paleogeographic coherence and further supports the synchronous nature of their emplacement. The calculated paleopole is plotted on the synthetic APWP for India (Figure 11), based on the GAWPWaP model of Torsvik et al. (2012), with the DSP (Vandamme et al., 1991) included for reference and the composite stratigraphic column of the Deccan Traps from the Western Ghats is shown in Figure 12.

The paleolatitudes depict the northward drift of the Indian Plate during the late Cretaceous, as documented by Klootwijk (1976, Figure 13). Specifically, the paleolatitudinal positions of the Pachmarhi dyke at 22.75°S and the Nandurbar-Dhule (N-D) dyke at 25.4°S (Das & Mallik, 2021) demonstrate a minimal variation in latitude and hence time between the emplacement of eastern (Pachmarhi dykes) and western (N-D dykes) segments of the Narmada-Satpura-Tapi dyke swarm (Figure 13). This consistency supports the hypothesis of near-synchronous dyke emplacement across the region.

Das and Mallik (2021), using paleolatitudinal differences and the spatial relationships between the Deccan lava flows and their associated dykes, estimated the Indian Plate's velocity at approximately 25 ± 8 cm/year during the Late stages of Deccan volcanism, based on their study of the Nandurbar-Dhule dyke swarm. This estimate closely aligns with earlier reported plate velocities of 20–25 cm/year for this interval (Cande & Stegman, 2011; Chatterjee et al., 2013; Copley et al., 2010), supporting the view that the Indian Plate maintained a phase of rapid northward drift through the final stages of Deccan volcanism. However, we were unable to calculate the plate

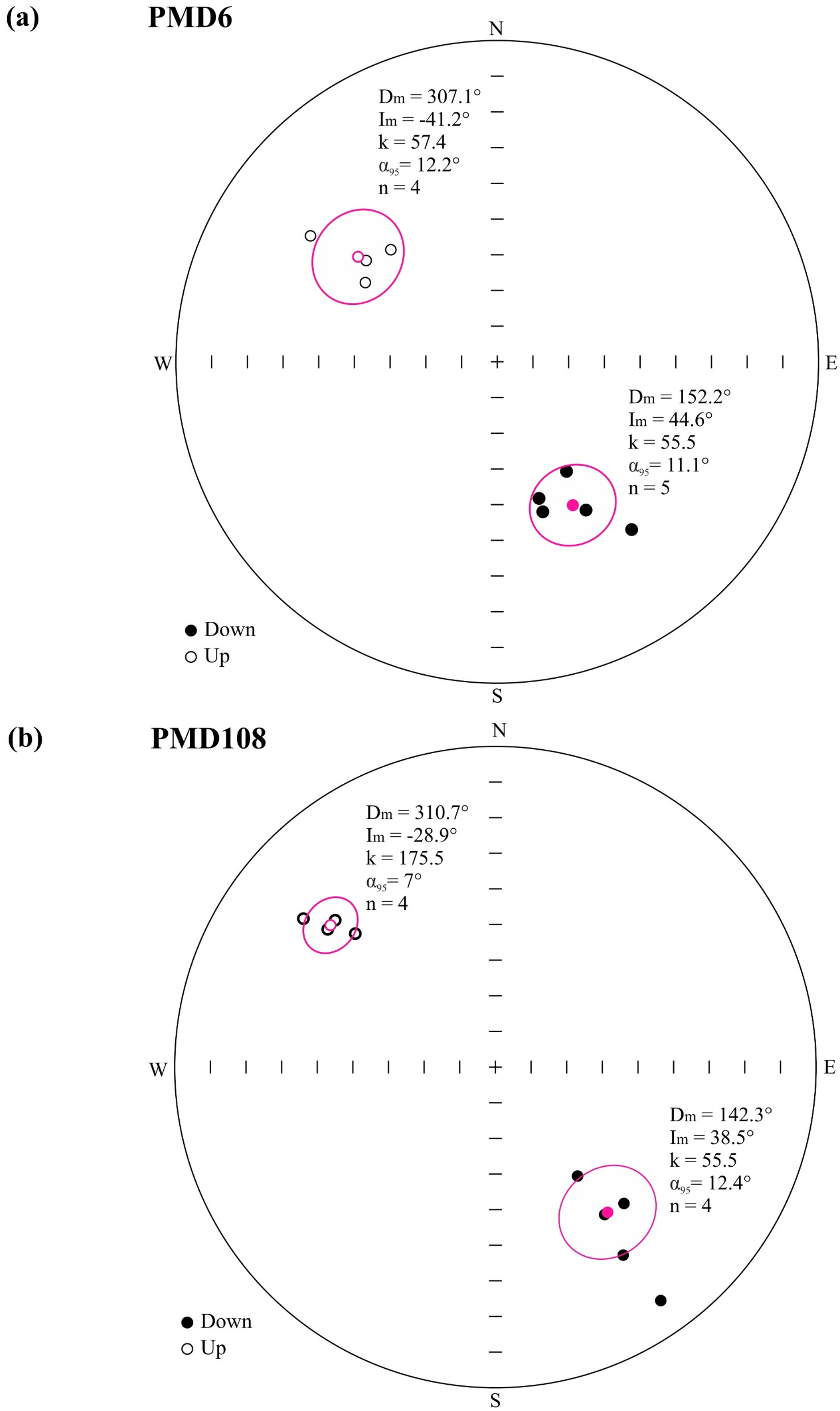

Figure 10. Equal area projections of site-mean characteristic remanent magnetization (ChRM) directions for Pachmarhi dykes PMD6 (Figure 3a) and PMD108, showing magnetic polarity reversal.

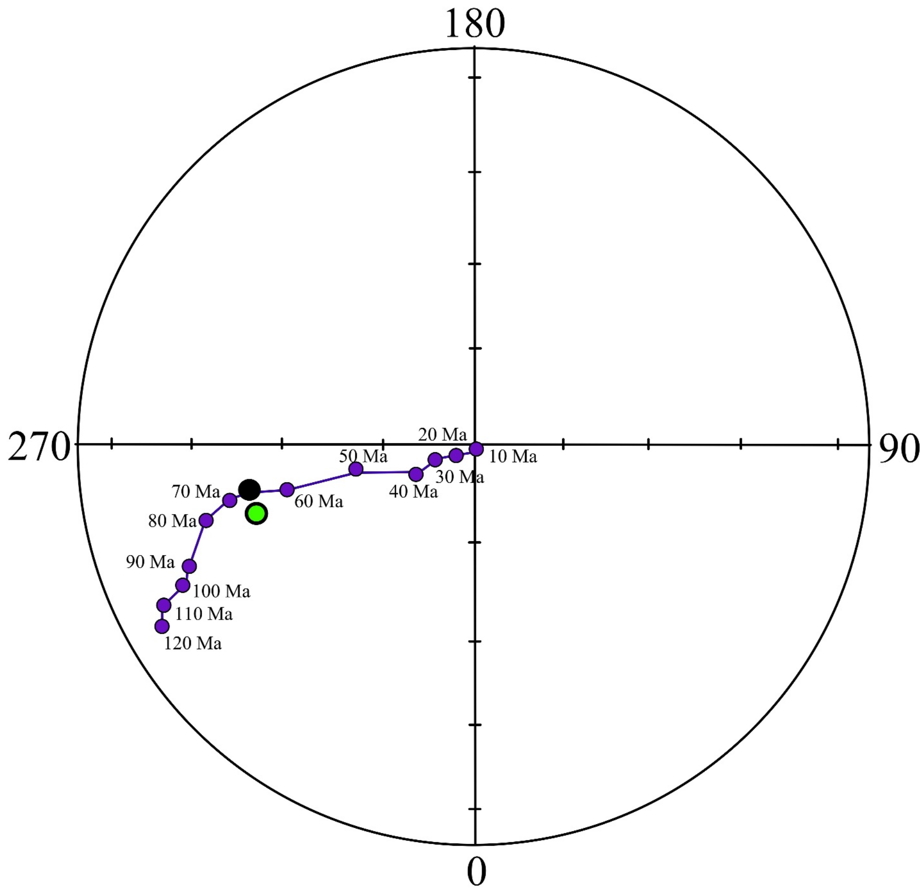

Figure 11. The mean paleomagnetic pole for the Deccan Traps plotted alongside the synthetic APWP for India based on the GAWPWaP model (Torsvik et al., 2012). The green circle marks the average pole position of the N–S–T dykes (37.8°N, 286.1°E, [73.9°W]), calculated from the results of this study combined with data from the Nandurbar–Dhule dykes (Das & Mallik, 2021) (38.3°N, 280.7°E [79.3°W]). The black solid circle denotes the mean Deccan Superpole at 36.9°N, 78.7°W, as reported by Vandamme et al. (1991).

velocity during the emplacement of the Pachmarhi dyke swarm because its host rock is Gondwana sandstone, unlike the Nandurbar-Dhule dykes, which are hosted in Deccan basalt.

This high velocity represents the continuation of an anomalous acceleration of the Indian plate that began around 90 Ma (Bardintzeff et al., 2010; Torsvik et al., 2000). This long-lived rapid drift phase has been linked to a combination of ridge push at the Central Indian Ridge and lithospheric weakening caused by extensive volcanic activity (Jagoutz et al., 2015). Plate motion reconstructions indicate that this rapid movement persisted until the Indian plate approached Trans-Tethyan Subduction Zone (TTSZ), where convergence slowed between ~50 and 55 Ma (Martin et al., 2020). Geological evidence from ophiolites and intra-oceanic arcs, such as the Kohistan–Ladakh arc, provides key constraints on the timing and mechanics of this collision (Beck et al., 1996; Tapponnier et al., 1981). Recent geochronological studies, including dating of the Khardung volcanics in the Kohistan–Ladakh arc, place intra-oceanic arc activity between ~62 and 66 Ma at paleolatitudes near 8°N (Martin et al., 2020), offering an important framework for linking plate velocities to paleogeographic reconstructions.

The plate velocity of ~25 ± 8 cm/year inferred from the Nandurbar-Dhule dykes indicates that India likely reached the TTSZ between ~51 and 54 Ma, which aligns with reconstructions of the initial India–TTSZ collision. This elevated plate velocity reflects the continuation of the long-lived anomalous acceleration that began around 90 Ma, persisted through late-stage Deccan magmatism (~66–63 Ma), and gradually decelerated as the Indian Plate approached the TTSZ (~50–55 Ma). Although the Réunion plume may have contributed to this higher plate speed (e.g., Cande & Stegman, 2011), plume activity alone cannot fully account for the magnitude or duration of the observed acceleration (Gurnis & Torsvik, 1994). Instead, a combination of factors, including ridge push, slab

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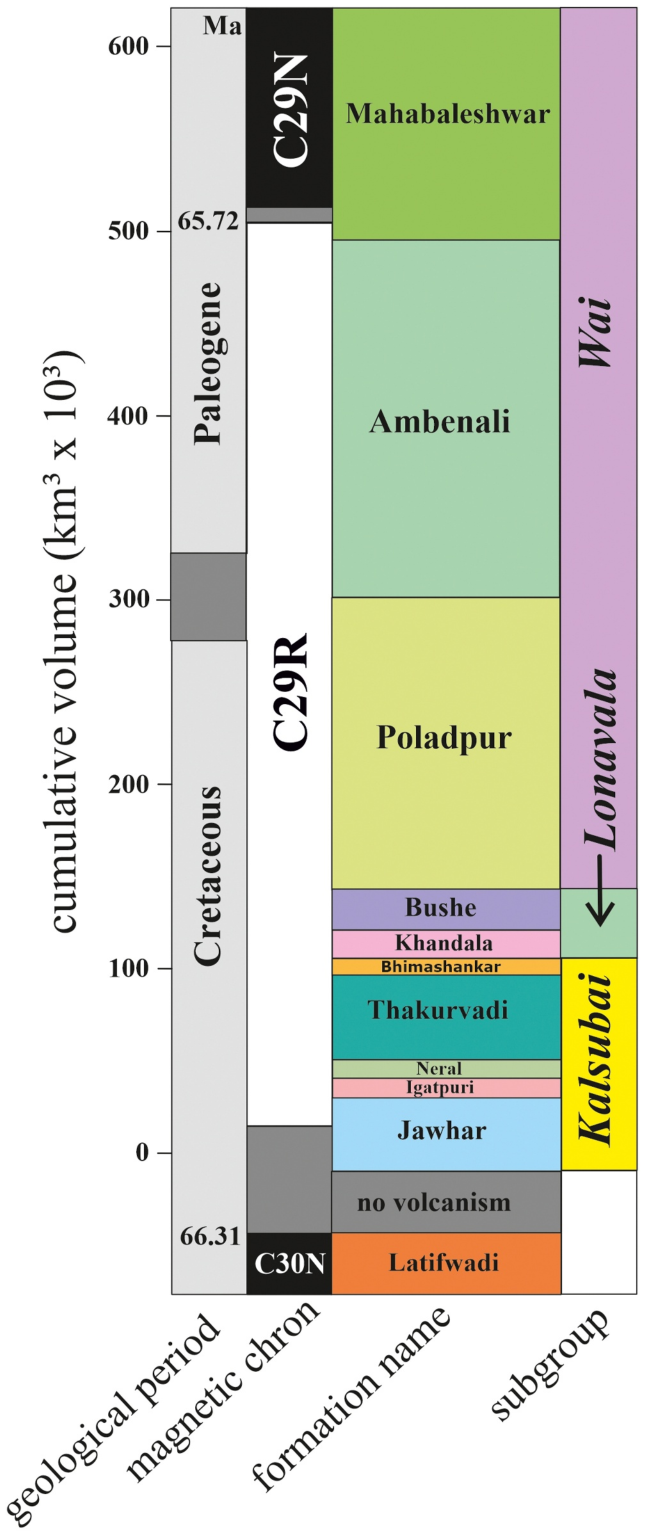

Figure 12. The composite stratigraphic column of the Deccan Traps from the Western Ghats, illustrating cumulative eruptive volume on the left according to the published volume model. Figure modified after Renne et al. (2015), Richards et al. (2015), Schoene et al. (2015, 2021), and Sprain et al. (2019).

pull at the TTSZ, and lithosphere-asthenosphere decoupling associated with Deccan magmatism, likely drove this phase of rapid northward motion (Jagoutz et al., 2015; Kumar et al., 2007).

The rapid northward drift of the Indian Plate likely made the asthenospheric drag stronger, increasing tensional stresses in the crust along the Narmada–Son Lineament (NSL), an ancient, weaker tectonic zone. This Precambrian tectonic zone, characterized by crustal weakness, underwent extensional fracturing, enabling tholeiitic magmas to ascend and form the Narmada-Satpura-Tapi dyke swarm, including the Nandurbar-Dhule and Pachmarhi dyke swarms (e.g., Das & Mallik, 2021). Together, these processes reflect the coupled effects of plume activity, ridge dynamics, and tectonic stresses that governed Indian Plate motion and magmatism during the Cretaceous–Paleogene interval (e.g., Ju et al., 2013).

5.1. Structural Controls and Episodic Magma Emplacement

Dual polarity in the Pachmarhi Deccan dykes, especially in PMD6 and PMD108, offers important new information about the late-stage development of Deccan volcanism during the C29R–C29N transition timeframe. Both normal and reversed polarities are present in PMD6, indicating that a geomagnetic reversal was captured by several magma injections that took place over a long period of time through a single conduit (Figures 3a and 14). This observation aligns with the findings of Liebke et al. (2012), who demonstrated that antipodal remanence directions in basaltic-andesitic dykes from the Lhasa Block were a result of distinct geomagnetic field polarities rather than self-reversal. Their modeling of cooling times showed that these polarity variations were preserved by rapid cooling, suggesting that emplacement was caused by multiple magma pulses rather than a single intrusion event. Furthermore, Bons et al. (2004) proposed that magma transport frequently occurs in discrete batches, while Gudmundsson (1984, 1995) described basaltic dykes forming through multiple injections separated by intervals of up to 1,000 years. This episodic emplacement model effectively explains why PMD6 records successive magma pulses spanning a geomagnetic reversal, indicating its prolonged role as an active magmatic pathway. Figure 14 shows this process by showing a dyke created by successive magma pulses inside a single fracture in both plan and cross-section. If Earth's magnetic field reverses during cooling, a dyke like PMD6 may record a polarity transition if it experiences prolonged intrusion. The later flow units in this scenario recorded the more recent normal polarity (C29N in Deccan), whereas the earlier ones maintained the older reverse polarity (C29R in Deccan). PMD6 was reactivated several times rather than being placed in a single event because both polarities were present in the same dyke. The existing fracture was utilized by each new magma injection, creating unique flow units that cooled under various geomagnetic circumstances. This episodic style of magma emplacement reflects the dynamic and long-lived nature of dyke systems within the Deccan volcanic province, revealing complex interactions between magma supply, fracture reactivation, and changes in the geomagnetic field.

An understanding of the relationship between fracture dynamics and magma emplacement is crucially provided by the structural evolution of PMD108 and the nearby dykes. The formation of an ~E-W-oriented dyke that records only reverse polarity was initially made possible by two east-west (~E-W) trending fractures that allowed magma intrusion. A new channel for magma transport was created by a north-south (~N-S) trending fracture that

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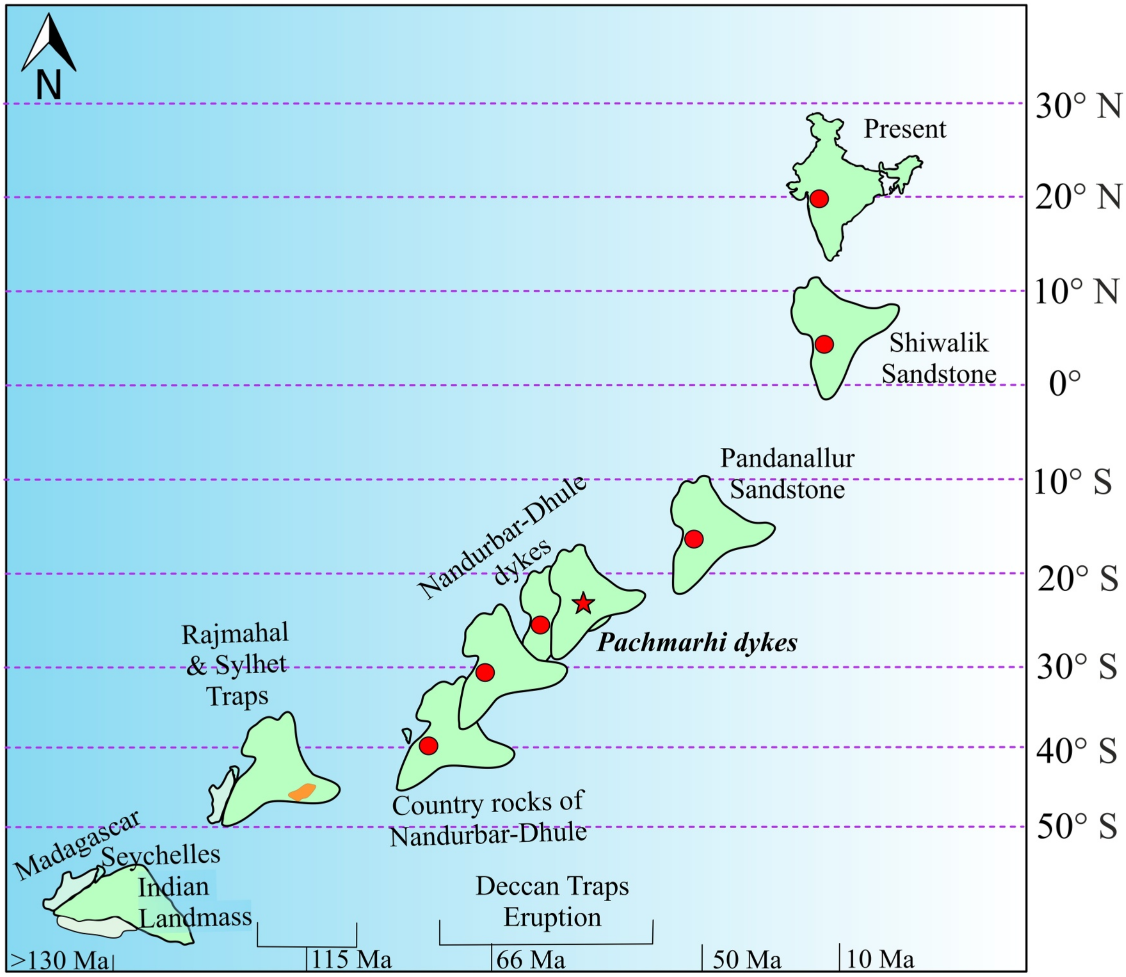

Figure 13. The red circle represents the paleolatitudinal positions of the Indian Plate across various time periods following the Deccan eruptions, while the red star indicates its position during the emplacement of the Pachmarhi dykes (modified after Klootwijk, 1976; Das & Mallik, 2021).

developed as a result of a subsequent shift in the local stress field. In contrast to the E-W dyke, the N-S dyke displays both normal and reverse polarity, indicating that it was positioned via a geomagnetic reversal that involved several magma injections. Older reverse polarity (C29R) was captured by the earlier intrusions, whereas the later ones showed a shift in the Earth's magnetic field to normal polarity (C29N).

The structural complexity of dyke emplacement is best illustrated by the formation of PMD108, which trended 140°. Its orientation indicates that it was formed by the interaction and linkage of propagating fractures, as it is situated between PMD1 (52°) and PMD2 (80°). Because PMD108 has mixed polarity, it suggests that magma pulses exploited an evolving fracture system, with each intrusion documenting the geomagnetic field that prevailed at the time of cooling. These observations are consistent with dyke propagation models, which preserve multiple magnetic signatures within a single dyke structure by creating pathways for episodic magma transport through fracture coalescence (Figure 15). The shift from ~E-W to ~N-S fracturing points to a local stress regime reorientation, most likely caused by changes in magmatic pressure or more extensive regional tectonic influences. The observed polarity transitions were caused by a stress realignment that not only regulated the formation of new fractures but also affected the timing of magma emplacement.

5.2. Emplacement History of Deccan Intrusives

The emplacement ages of dykes provide key insights into the overall timing and dynamics of Deccan volcanism. Courtillot et al. (1986) established that the Deccan Traps formed in three magnetic chrons (C30N, C29R, and C29N). During the reversed C29R chron, which occurred close to the Cretaceous–Palaeogene (K–Pg) boundary,

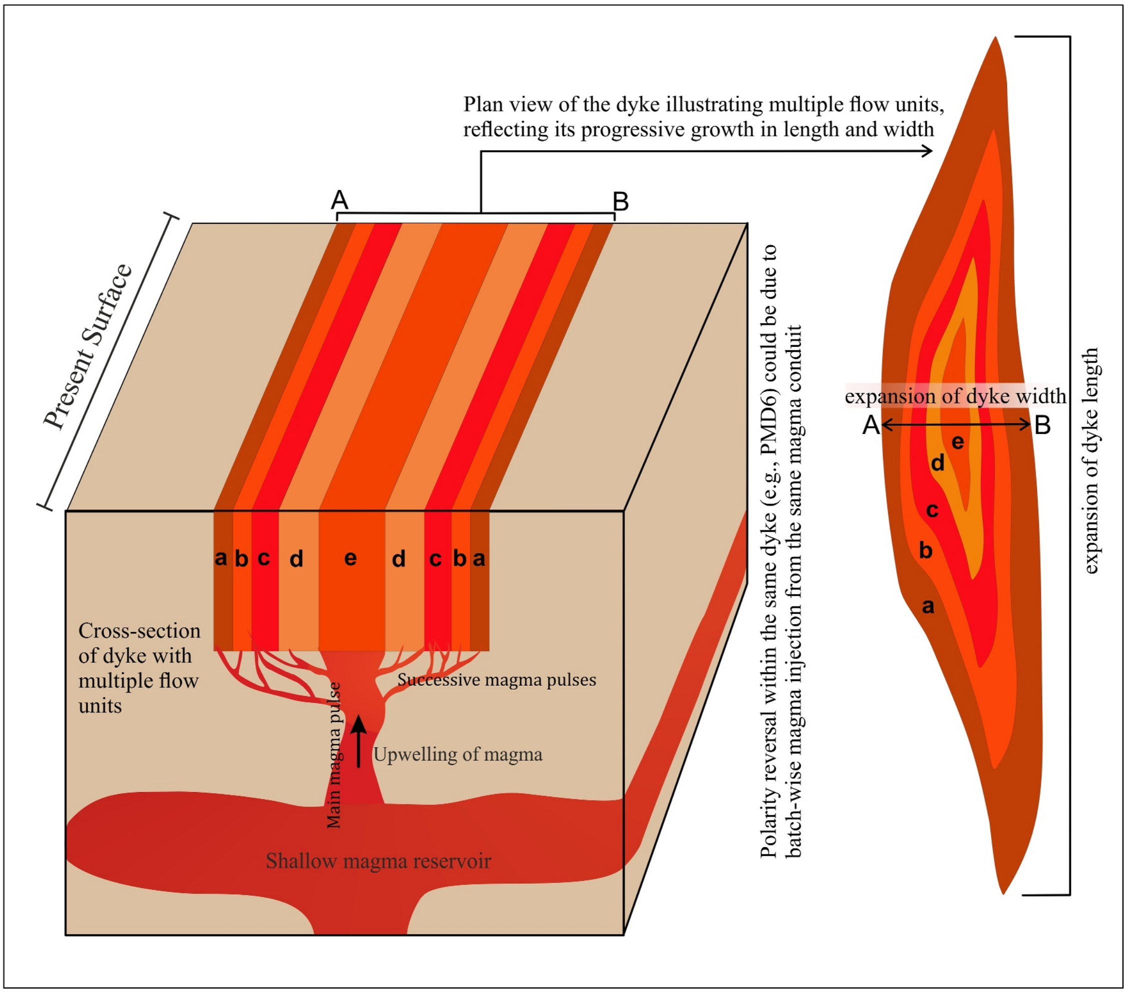

Figure 14. The figure illustrates the cross-section and plan view of a dyke PMD6, formed through main and successive magma pulses injected into a single fracture from a shallow magma reservoir. This dyke has recorded a polarity transition of the Earth's magnetic field during its cooling, with earlier and later flow units capturing different magnetic polarities. Although it is unclear whether the transition is normal-to-reverse or reverse-to-normal, the preserved pattern reflects cooling through a geomagnetic reversal.

about 80% of the lava flow was emplaced. Initial estimates suggested that Deccan volcanic activity persisted for less than one million years. However, subsequent studies incorporating geochronological, paleontological and paleomagnetic data (Chenet et al., 2007; Keller et al., 2008) refined this chronology, identifying three distinct eruptive phases: an initial brief episode around 67.5 ± 1 Ma near the C30R–C30N reversal, followed by a 2.5-million-year quiescence and two major eruptive pulses occurring at ~65 ± 1 Ma—one within C29R and the other near the C29R–C29N transition.

According to Chenet et al. (2008), Deccan volcanism was believed to have happened in three distinct eruptive phases: Phase 1 during Chron C30N, a significant Phase 2 during Chron C29R, and a final Phase 3 during Chron C29N. Long intervals between eruptions were suggested by this phase-based model. However, recent geochronological and paleomagnetic studies significantly revised this perspective and suggest that 80% of the Deccan basalts erupted in a short span (~68-64 Ma) straddling the Cretaceous -Paleogene Boundary (~66 Ma), followed by a smaller phase till ~62 Ma (Eddy et al., 2020; Kale & Pande, 2022; Schoene et al., 2015, 2019, 2021; Sprain et al., 2019; Tholt et al., 2023). Sprain et al. (2019) concluded that the main phase of Deccan volcanism within the Western Ghats region did not erupt in three discrete large pulses, as previously suggested, but rather erupted over a quasi-continuous period of approximately 991,000 years.

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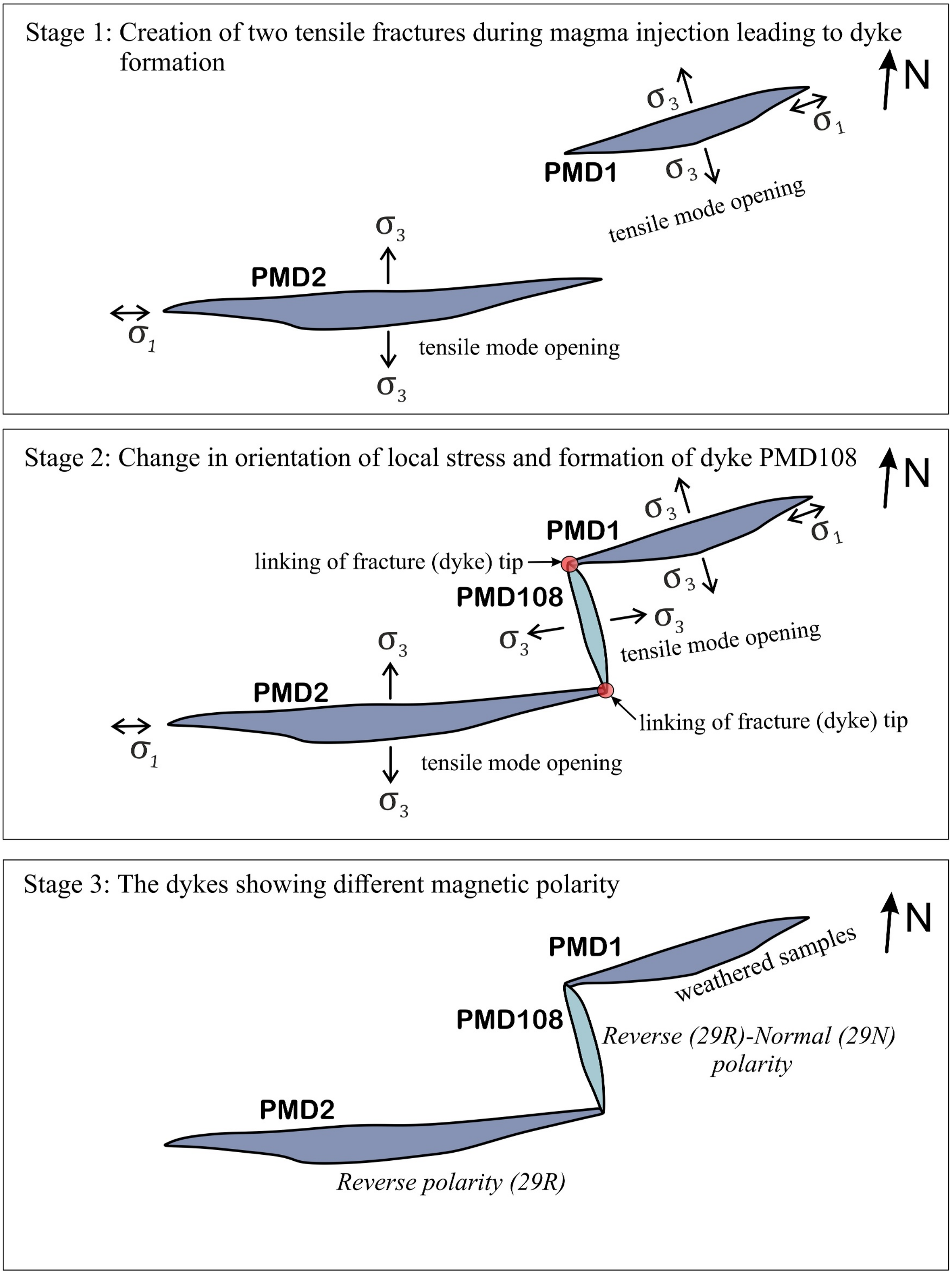

Figure 15. The figure depicts a model of dyke propagation, where fracture coalescence creates pathways for episodic magma transport, preserving multiple magnetic signatures due to reorientation of the local stress regime. Here, PMD2 depicts reverse polarity, and PMD108 shows both the reverse and normal polarity.

High-precision U-Pb zircon and $^{40}Ar/^{39}Ar$ dating suggest that Deccan volcanism consisted of overlapping eruptive intervals rather than separate phases (Eddy et al., 2020; Schoene et al., 2015). Schoene et al. (2019) improved this conception by highlighting four high-volume eruptive pulses in the Deccan rather than three.

According to their findings, a major pulse began just tens of thousands of years before the bolide impact, with peak eruption rates occurring both before and after the K-Pg boundary. Renne et al. (2015) and Richards et al. (2015) interpret the $^{40}Ar/^{39}Ar$ data set as indicating an increase in eruption rates during the Chicxulub impact, despite the fact that it did not identify such pulses. This multi-pulsed eruptive history demonstrates the Deccan volcanism's long-lasting environmental effects and possible contribution to the end-Cretaceous mass extinction. Schoene et al. (2021) in their studies have highlighted the importance of addressing systematic uncertainties between dating methods and suggested the regional stratigraphic correlations for accurate eruption models. The main phase of Deccan Traps eruptions began near the C30N–C29R magnetic reversal and continued for roughly 700–800 kyr (Schoene et al., 2019, 2021; Sprain et al., 2019, Figure 12). The U−Pb zircon geochronology by Schoene et al. (2019) indicated four major eruptive pulses occurring both before and after the Cretaceous–Paleogene boundary (KPB), while the $^{40}Ar/^{39}Ar$ ages by Sprain et al. (2019) did not support this interpretation. The $^{40}Ar/^{39}Ar$ data set showed more constant eruption rates with substantial uncertainties throughout the duration of the eruptions, providing no clear evidence for the pulses identified by the U−Pb data.

Geochronological constraints for the Pachmarhi dykes are still lacking, thus leaving it uncertain which major eruptive pulse they correspond to. Our study shows that these dykes exhibit both normal and reversed polarity, suggesting emplacement across at least one magnetic reversal interval. Geochemical evidence (Sheth et al., 2009) tentatively places them near the C29R–C29N transition, but this remains uncertain without direct dating. However, high-precision U–Pb zircon geochronology from the Malwa Plateau provides a solid framework. Eddy et al. (2020) show that the lower Malwa Plateau basalts correspond with the initial phase of Deccan volcanism, and they constrain the lower magnetic reversal to 66.275 Ma to 66.256 Ma, confirming its association with the C30N/C29R reversal. The result by Eddy et al. (2020) is consistent with several previous estimates, including a U–Pb zircon date from a transitional polarity basalt flow in the Western Ghats (Schoene et al., 2019) and high-precision $^{40}Ar/^{39}Ar$ sanidine dates from interbedded tuffs in the Hell Creek Formation, Montana (Sprain et al., 2018).

High-precision U–Pb zircon dates from Eddy et al. (2020) and Schoene et al. (2019) also indicate that the Narmada Formation erupted slightly earlier (~60 kyr) than the Jawhar Formation, extending the onset of Deccan magmatism to at least 66.358 Ma. Eddy et al. (2020) further suggested that the first eruptive pulse was more voluminous in the north, where it intruded and erupted through organic-rich sediments of the Narmada–Tapti rift basin, potentially enhancing volatile release.

Magnetostratigraphic data place the younger Mhow, Satpura, and Singachori Formations, between 66.137 and 65.631 Ma, broadly synchronous with the second to fourth eruptive pulses of the Western Ghats. The Singachori Formation likely correlates with the Mahabaleshwar Formation, as both record the C29R/C29N reversal (Eddy et al., 2020). Eddy et al. (2020) therefore concluded that these three Malwa Plateau formations correspond to Schoene et al. (2019) second-fourth eruptive pulses and are temporally equivalent to the Poladpur, Ambenali, and Mahabaleshwar Formations, respectively. Tholt et al. (2023) further showed that the Malwa Plateau eruptions peaked between 66.4 and 66.3 Ma during the Late Maastrichtian Warming episode, having started around 400 kyr earlier than the Western Ghats lava at 66.8 Ma. The lavas, which indicate an earlier, colder stage of the Deccan plume, originated from lower degrees of partial melting at cooler mantle temperatures and shallower depths. These studies have highlighted the geochemical and chronological variations in Deccan volcanism.

Within the above-mentioned context, the Pachmarhi dyke swarm, which marks the eastern end of the Narmada-Satpura dyke swarms, may have recorded a similar sequence of magnetic reversal. Their dual polarity may be consistent with episodic magmatism spanning the C30N–C29N interval, encompassing both the C30N/C29R and C29R/C29N reversals (Figure 12), similar to patterns documented in the Malwa Plateau, or just a lower (C30N/C29R) or upper (C29R/C29N) magnetic reversal (Eddy et al., 2020; Tholt et al., 2023). Field observations reflect this interpretation. PMD108, a structurally complex dyke resulting from an interaction of propagating fractures, shows mixed polarity, whereas PMD2 exhibits consistent reverse polarity. Collectively, these intrusions suggest that magma ascended through a developing fracture network during a geomagnetic field transition, likely enabling emplacement over the C29R–C29N boundary (Figure 15). Yet this conclusion remains uncertain due to the lack of precise geochronological constraints. Although Pachmarhi intrusions such as dyke PMD2 and sill PMS1 (Figure 1c) show geochemical affinities with formations like Poladpur, Ambenali, and Mahabaleshwar formations of the upper Wai Subgroup, their distinct isotopic signatures suggest they are not direct feeders for these specific lava flows (Sheth et al., 2009). Instead, they represent the results of complex magmatic processes, emphasizing the difficulties in linking intrusions with distant lava flows across a large igneous province like the Deccan Traps (Sheth et al., 2009). Overall, these findings underline the necessity for accurate geochronology to further constrain paleomagnetic interpretations by supporting the hypothesis that dyke emplacement took place in discrete, time-separated pulses.

These findings underscore the significant influence of structural controls on dyke emplacement and the preservation of geomagnetic reversals in volcanic provinces. The merging of dyke tips, as observed in PMD108 and the repeated magma injections in PMD6, demonstrated how local stress variations influence magma pathways and polarity recording. This aligns with global studies of mafic dykes, such as those in Iceland (Gudmundsson, 1995), where composite dykes formed through successive injections over extended timescales. Ultimately, the study of these Pachmarhi dykes contributes to a more comprehensive understanding of the structural and magmatic processes governing the evolution of the Deccan Volcanic Province and its broader geodynamic implications for flood basalt provinces worldwide.

6. Conclusion

The Pachmarhi dykes exhibit normal and reverse polarity ChRM directions at $D_m = 329.3°$ and $I_m = -40.3°$ ($k = 80.1$, $\alpha_{95} = 8.6°$, $n = 31$) and $D_m = 154.4°$ and $I_m = 39.5°$ ($k = 122$, $\alpha_{95} = 5.5°$, $n = 69$), respectively, with an overall paleomagnetic pole at 37°N and 68.6°W. This paleopole closely aligns with the Nandurbar-Dhule (N-D) dykes (38.3°N, 79.3°W), averaging to 37.8°N, 73.9°W, consistent with the DSP (36.96°N, 78.7°W). The estimated paleolatitude for the Pachmarhi dykes is 22.75°S, with minimal variation compared to the N-D dykes (25.4°S), highlighting synchronous emplacement across the Narmada-Satpura-Tapi dyke swarm during late-stage Deccan volcanism.

Two of the studied Pachmarhi dykes exhibit both normal and reverse polarity, suggesting that magma was emplaced in multiple pulses rather than a single intrusion. This suggests that these dykes continued to function as active conduits for a considerable amount of time, possibly even during a geomagnetic reversal. These findings are consistent with an episodic model of dyke emplacement, in which alternating magma injections document shifting geomagnetic field orientations after cooling. Future research should combine high-precision geochronology and advanced rock-magnetic analyses to further constrain the timing and dynamics of these intrusions, giving a more comprehensive picture of the Deccan province's magma transport processes.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Data Availability Statement

All paleomagnetic data produced in this study are archived in the Magnetics Information Consortium (MagIC) database (Shukla et al., 2025).

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